

Data Processing : Stream & Batch processing



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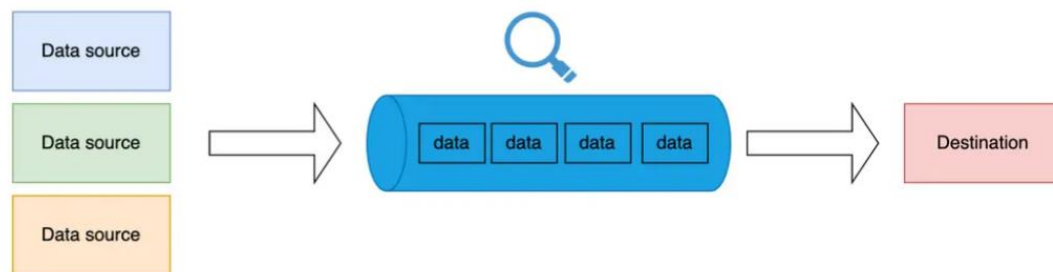
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Stream Processing

Incoming data is processed in real time.
Processing output is within seconds or minutes.

For getting Insights / alerts / notifications.

Requires lower processing power but continuous availability.
Can be expensive



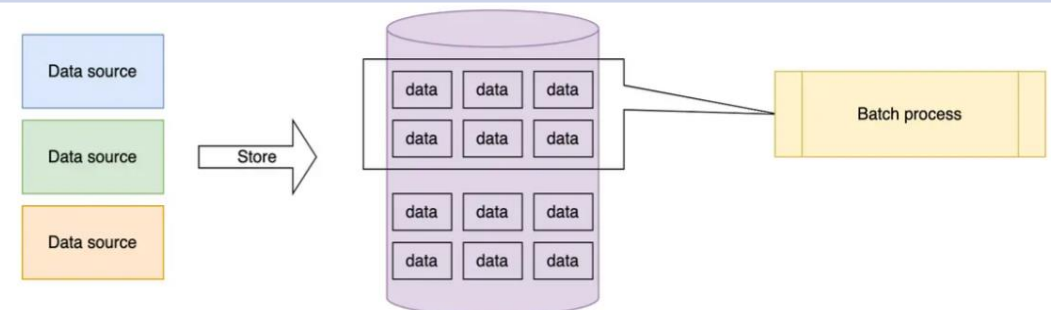
Stream Processing

Batch Processing

Data collected over a period of time is processed together.
Processing may take minutes, hours even days

For trends, correlations, insights, historical analysis

Requires heavy processing power for shorter duration.
Can be very cost effective.



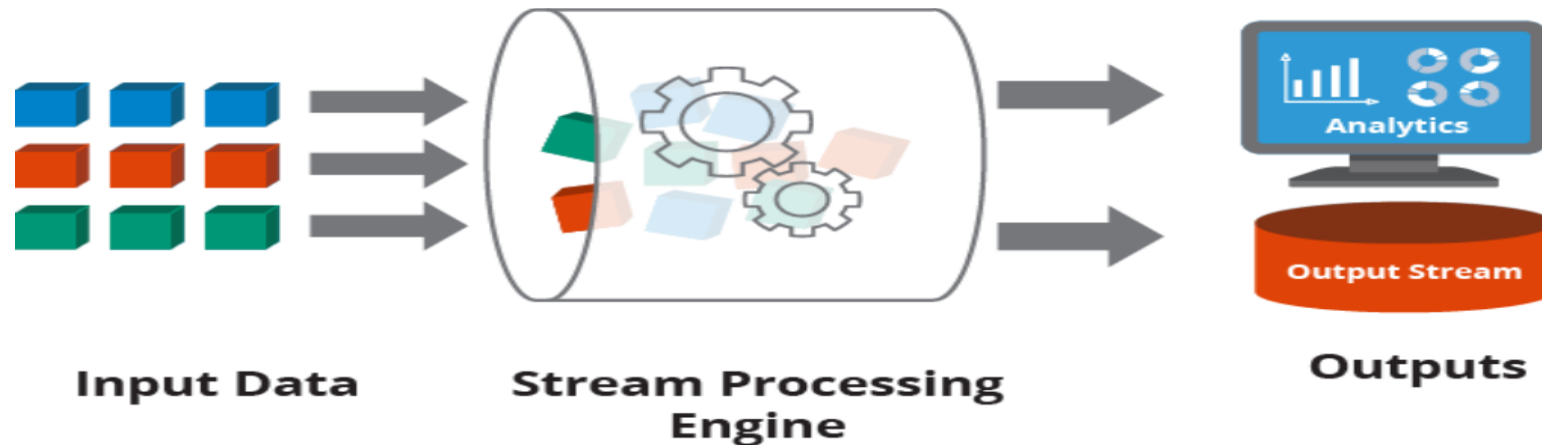
Batch Processing

Stream processing



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Input Data from multiple sources

- IOT Sensors
- Clickstreams
- Mobile apps
- Application Logs

Stream Processing Engine processes incoming data in real time

- Aggregations
- Analytics,
- Transformations,
- Enrichment
- ingestion

Output data is delivered to a streaming analytics application and added to the output stream

Stream Processing Use Cases

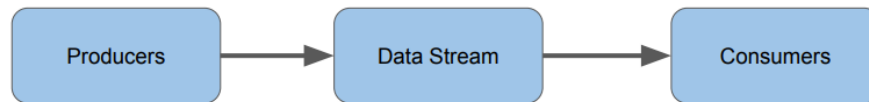
- ▶ Hospitals
- ▶ Traffic Management
- ▶ Safety and Security Surveillance
- ▶ Energy Management
- ▶ Real time Financial Fraud detection
- ▶ Digital Marketing
- ▶ Autonomous Driving Cars
- ▶ High Frequency Trading
- ▶ Social Media Analysis
- ▶ Log and Infrastructure monitoring

Stream Processing - How it works ?

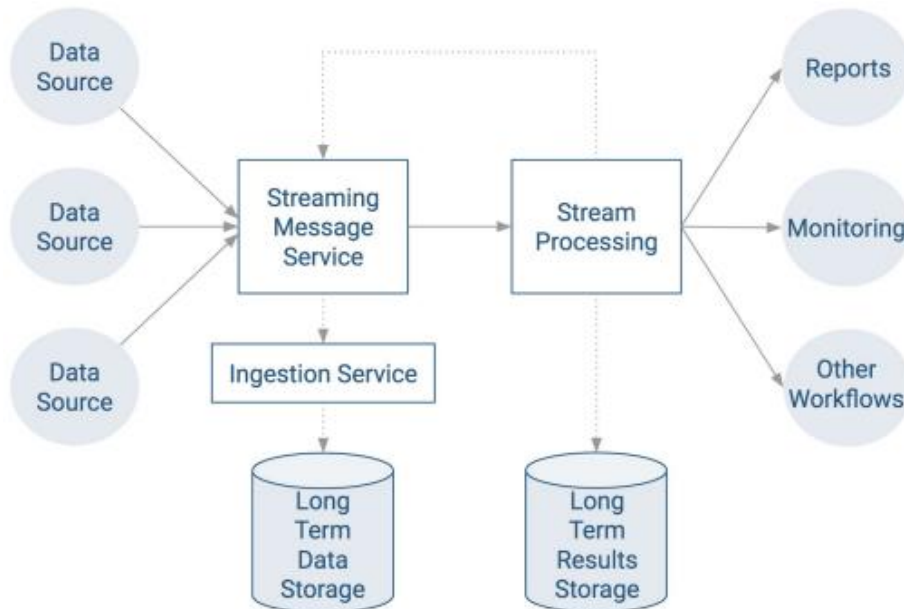


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Data Streaming Pipelines Architecture



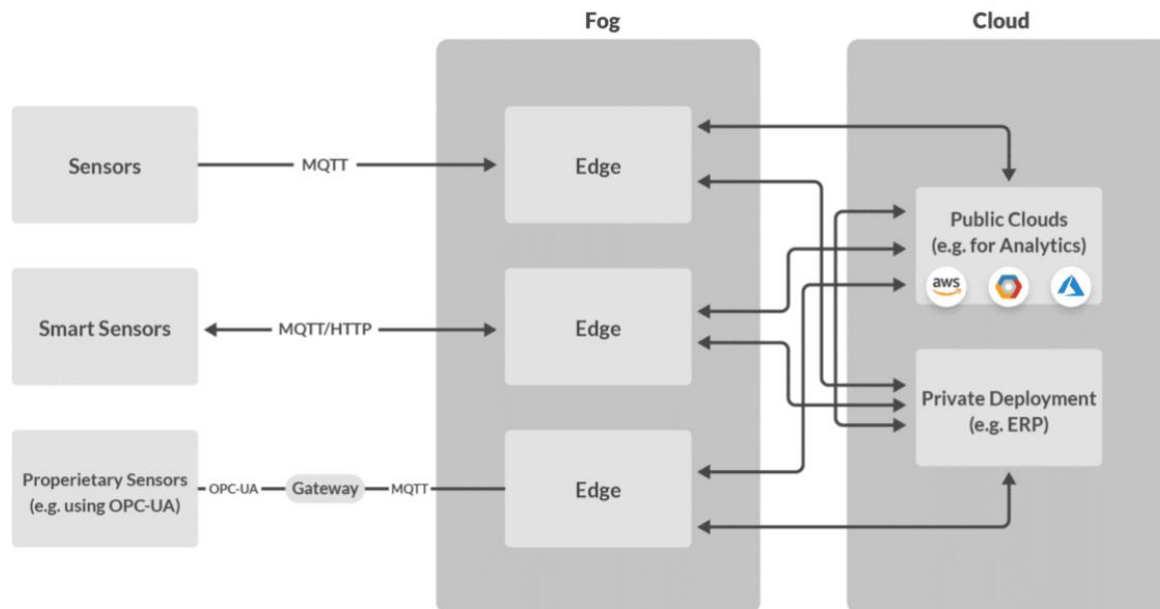
- Follows the pub-sub model
 - **Producers** generate data send it to a stream framework
 - **Consumer** consumes data
- Real Time Processing
 - Alerts, micro-batch processing and transformations
 - Pipelined workflows can involve transformations across multiple immediate streams
 - Processed data are channelled for respective usage and long terms storage endpoint
- Data flow scaling is handled using **partitions or shards**. Handles buffered data storage, error retries and scale as the data throughput increases.
- **Kafka, AWS Kinesis, Azure Stream Analytics** are some leading stream processing Engine.

Stream Processing in IoT



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IoT Challenge

- Heterogenous devices generating large amount of data (V4 – Velocity, Volume, Varsity, Veracity). Communication and Bandwidth challenge
- Stream Processing Use Case in IoT
 - Monitor and Aggregate data from sensors
 - Detect events of importance and trigger action
 - Validate sensor data and remove outliers
 - Enable real-time predictions and required optimizations
- Stream processing can be done at Gateway, Edge Fog, Cloud

Architecture and Design Considerations for Stream Processing

- **High availability** to handle continuous incoming high-frequency data
- **Scalability** using partitioning semantics to handle big data streaming with periodic or sudden data bursts
- **Security** -Encryption during transfer for various sensitive use cases
- **Buffering** large amounts of data to account for delays on data bursts
- **Queue** ordering, delivery confirmations, error handling etc.

Stream Processing : Tools

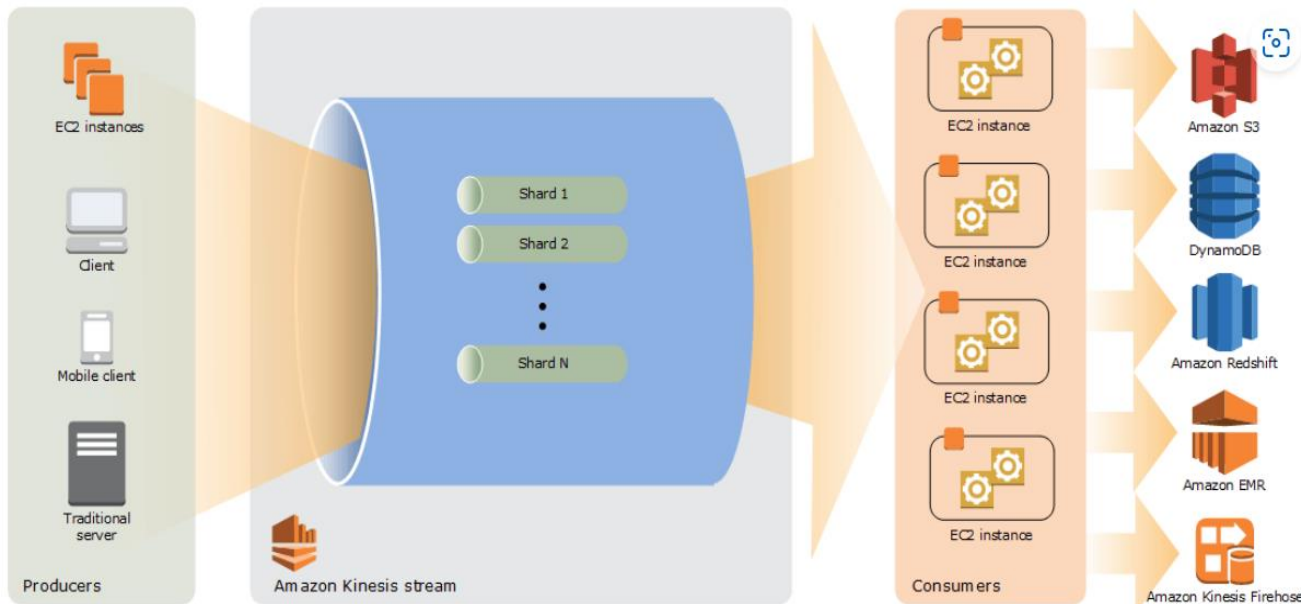
AWS Kinesis	• Fully managed service from AWS to process streaming data at scale. • Azure Stream Analytics and Google Cloud Stream
Kafka	• Open source distributed pub-sub messaging system. • Highly reliable and scalable • It's used for big data streams like Netflix's big data platform. • Integrates well with Spark. • Apache Storm and Flink are similar tools.
Spark Streaming	• Enables both high speed batch processing and micro-batch streaming and micro batch processing • A new paradigm called Structured streaming allows for streams to behave as unbounded tables with each stream data point acting as a table row.
AWS SQS Azure Storage Queue (ASQ)	• Provides scalable fully-managed queues for a decoupled pub-sub architecture • Supports encryption, retries, FIFO ordering, various queue types and tracking job completions
Kinesis Vs SQA	• Kinesis is preferred where real time processing is needed • SQS is more about Jobs where jobs are sent in background or processing are send to down dream applications

Stream Processing : AWS Kinesis



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- **Fully managed service** for real-time data processing.
 - Input from various Kinesis Data stream
 - Ingests and stores this data for processing
 - Data is processed / analysed by frameworks like Kinesis Data Analytics, Spark EMR, AWS EC2, AWS Lambda
 - Visualize output data using BI tools
- **Kinesis Streams** –pipes with Producers and Consumers
 - **Shards** :- sequence of data and is basic unit of scale.
 - Can scale infinitely in multiple of shard capacity only.
 - Re-sharding - dynamically added/removed
 - Single shard - 5 TPS for read upto 2 mb/sec OR
 - 1000 TPS for write upto max of 1 mb/sec
 - Record is unit of data stored in stream
 - Data record goes in a shard – 3 components –
 - sequence number
 - partition key
 - data itself (blob)

<https://aws.amazon.com/kinesis/>

Stream Processing

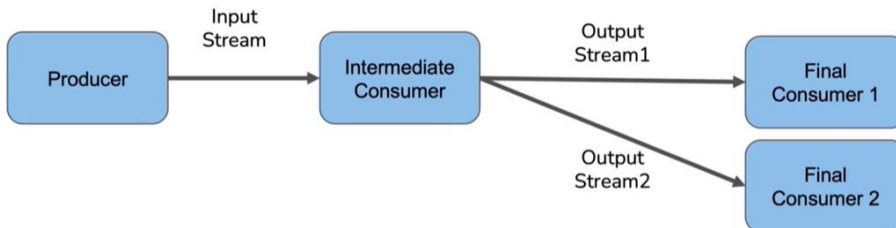
AWS Kinesis Pipes & Kinesis Firehose



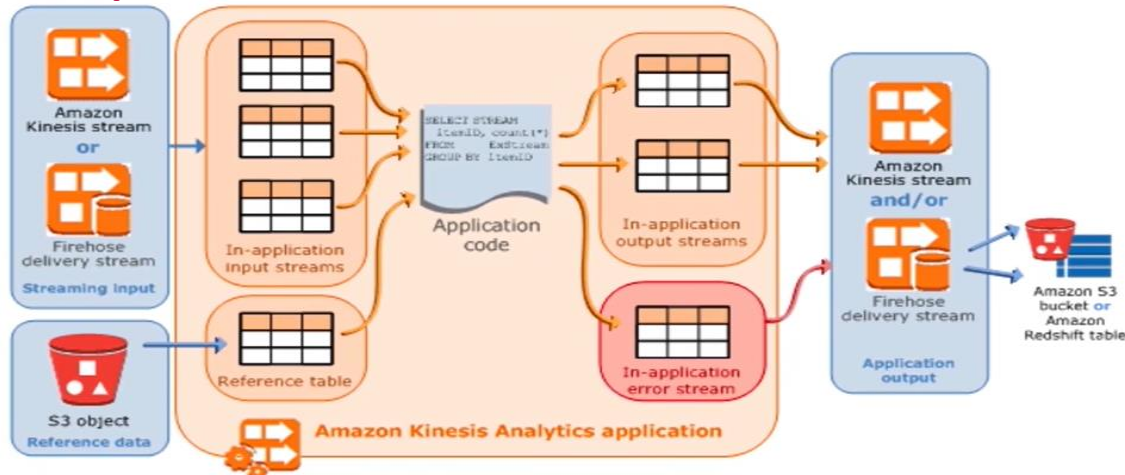
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Kinesis Pipes and Firehose are two Amazon Web Services (AWS) offerings that help process and stream large amounts of data



Multiple Stream Scenario



Kinesis Pipes

- Used to split a streams of data into multiple processing pathways. By leveraging Kinesis pipes, you can build sophisticated data processing workflows.
- Routes data to different processors, such as AWS Lambda functions, Amazon Elastic MapReduce (EMR), or Amazon Redshift.
- Kinesis pipes can have multiple consumer at each stage.
- Input stream is processed and transformed to various output streams.
- Consumers getting pre-processed transformed data for final processing.
- Transformed streams can be aggregated and stored for future analysis.

Kinesis Firehose

- Used for delivering streaming data to destinations such as S3 and Redshift or Elasticsearch without any additional processing
- Scales to GB per second of streaming data with auto scaling. In contrast Kinesis Streams requires manual provisioning of additional capacity.
- Data can be ingested using - HTTPS, Kinesis Producer Library, Kinesis Client Library, and Kinesis Agent.
- Monitoring is available through Amazon Cloudwatch

Stream Processing : AWS Firehose and SQS



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- **Kinesis Firehose** – data ingestion (ETL) product from Kinesis.
 - Used to load streaming data into other Amazon services such as S3 and Redshift. Further you can load the streams into analysis tools like EMR, and Amazon Elasticsearch Service.
 - Scales to gigabytes of streaming data per second with auto scaling. In contrast Kinesis Streams requires manual provisioning of additional capacity.
 - Data can be ingested using - HTTPS, Kinesis Producer Library, Kinesis Client Library, and Kinesis Agent.
 - Only possible to stream data via Firehose to S3 and Redshift. Then the data can be copied to other services for further processing and analysis. Monitoring is available through Amazon Cloudwatch
- **AWS SQS (Simple Queue Service)** - Kinesis enables real-time stream processing of big data. SQS, on the other hand, is used as a message queue to store messages transmitted between distributed components.
 - SQS provides for messaging semantics so that application can track the successful completion of work items in a queue, and you can schedule a delay in messages of up to 15 minutes.
 - Unlike Kinesis Streams, SQS will scale automatically to meet application demand.
 - SQS has lower limits to the number of messages that can be read or written at one time compared to Kinesis, so applications using Kinesis can work with messages in larger batches than when using SQS.

Hands-On

AWS IoT Core, Kinesis Dynamo DB



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